AP PRECALCULUS

Unit 1 • Lesson 1.1 • Guided notes

Topic 1.1 • Change in Tandem

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Describe how output values change as input values increase.
- Identify intervals where a function is increasing or decreasing, and where its graph is concave up or concave down.
- Identify zeros and the sign of a function from a graph, a table or a formula.

Vocabulary. function, input, output, increasing, decreasing, rate of change, concave up, concave down, zero, x-intercept, positive, negative.

1. What is a function?

A **function** is a rule that assigns to each _____ exactly _____ output value. We write $y = f(x)$: x is the _____ and $f(x)$ is the _____. As the input values change, the output values change *in tandem*.

2. Increasing and decreasing

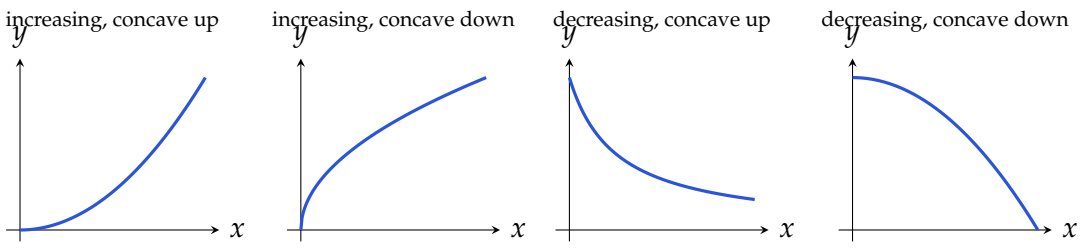
Key idea

On an interval, f is **increasing** if, as the input values increase, the output values always _____: if $a < b$ then $f(a)$ _____ $f(b)$.
f is **decreasing** if, as the input values increase, the output values always _____: if $a < b$ then $f(a)$ _____ $f(b)$.
Intervals of increase and decrease are written with _____ values.

3. Concavity: how is the function changing?

Key idea

The graph of f is **concave up** on an interval if the rate of change of f is _____: the graph bends _____.
The graph is **concave down** if the rate of change is _____: the graph bends _____.

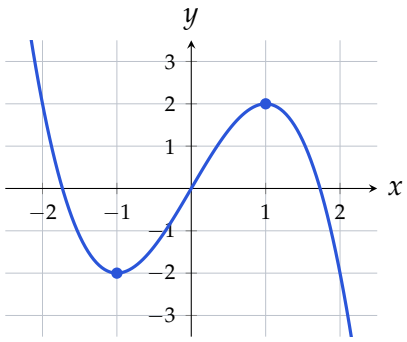


4. Zeros and sign

A **zero** of f is an input a such that _____. Graphically, the real zeros are the _____.
The function is **positive** where its graph is _____ the x-axis and **negative** where it is _____. A

function whose graph has no breaks can change sign only at a _____.

Example 1 • Reading a graph



The graph of $f(x) = -x^3 + 3x$ is shown. Find (a) the intervals where f is increasing and decreasing, (b) where the graph is concave up and concave down, (c) the zeros, (d) where $f(x) > 0$.

Example 2 • Reading a table

x	0	1	2	3	4	5
$g(x)$	-6	-1	2	3	2	-1

Describe how g changes. Where must g have zeros (assume the graph of g has no breaks)?

Example 3 • In context

The temperature T (°C) of a cup of coffee is $90, 70, 56, 46, 40$ at $t = 0, 5, 10, 15, 20$ minutes. Describe T.

Common mistakes

- Writing intervals with y-values instead of x-values.
- Confusing "decreasing" with "concave down". A graph can be increasing and concave down at the same time.

Check your understanding

1. Outputs at $x = 0, 1, 2, 3$ are $10, 7, 5, 4$. The function is _____ and its graph is concave _____.
2. The zeros of $f(x) = (x - 2)(x + 5)$ are _____.